Arnav Shah

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Education

Carnegie Mellon University, School of Computer Science — **Pittsburgh, PA**

B.S. Robotics, Minor in Creative Writing | ***GPA:*** *4.0/4.0* | ***SAT:*** *1600/1600* — *Expected May 2028*

- **Honors:** Dean's List with High Honors, 4 consecutive semesters (Fall 2024 to present); Peer Tutor for Imperative Programming & Linear Algebra
- **Coursework:** Multi-Robot Planning & Coordination^{G}, Mobile Robots^{G}, Space Robotics^{G}, Robot Learning^{G,*}, Computer Vision, Feedback Control, Robot Kinematics & Dynamics*, Intelligent Robot Systems*, Math Fundamentals of Robotics, Linear Algebra, Computer Systems, Theoretical CS — ^{G}graduate-level; *ongoing

Experience

Mujin Corp. — **Suwanee, GA**

Software Engineering Intern, Fleet Management — *May 2026 – Aug 2026*

- Shipped production planning features for 80+ robot AGV fleets in enterprise retail warehouses on a 1it/s replanning loop
- Extended fleet planner's cost function and graph representation to enable per-robot velocity constraints (loaded vs. unloaded) and dynamic slow-zone delineation for runtime hazards such as spills or equipment failures
- Detected and resolved cyclic dependencies arising from physical payload swaps by modifying task-graph dependency edges and execution priorities, preventing deadlock during paired exchanges

Carnegie Mellon Racing (Formula SAE) — **Pittsburgh, PA**

Driverless Controls & Vehicle Dynamics Engineer — *Aug 2025 – Present*

- **1st Overall, Formula Hybrid+Electric 2026; 1st Driverless Design, 4th Overall, Formula SAE Michigan**; deployed full Model Predictive Path Integral (MPPI) driverless stack achieving track speeds of up to 9 m/s
- Implemented top-k MPPI (top 2% trajectory averaging) as a temperature-tuning proxy, biasing rollouts toward high-reward actions and yielding a 1–2 m/s gain in both straight-line and cornering speed without loss of track adherence
- Corrected importance sampling weights in MPPI to account for hardware-clipped Brownian action distributions, restoring controller stability at actuation limits
- Re-engineered the perception-to-planner midline SVM (RBF kernel, retuned hyperparameters), halving the rate of corrupted midline outputs on noisy real-world cone detections (1/10 to 1/20 iterations)
- Upgrading vehicle dynamics model from kinematic bicycle to non-linear double-track with Pacejka tire forces, capturing slip and tire saturation for robust limit-handling

CMU Robot Intelligence Group (Prof. Jean Oh) & Momentum Lab (Prof. Jeff Ichnowski) — **Pittsburgh, PA**

Robot Learning Researcher, Open N-finger Omnidirectional End-effector (O-NOE); Bosch AI collaboration — *Apr 2025 – Present*

- Built the control and learning stack for an open-source modular tendon-actuated soft robotic finger, enabling researchers to assemble n-finger soft hands for scalable manipulation research
- Authored a SOFA physics simulation and Gymnasium environment wrapper for the gripper, enabling parallel RL training and integration with downstream Gaussian Splatting and design optimization workflows
- Trained 3-finger grasping policies with PPO and SAC under domain randomization and curriculum learning, achieving 80% cube-lift success rate in simulation at 8000+ environment steps per hour
- Built a sim-to-real action retargeting pipeline using ArUco-based motor-to-tendon-angle calibration with rotated force vectors to reconcile mismatched tendon mount geometries, transferring sim-trained policies to physical hardware

CMU MoonRanger Microrover (2029 Lunar South Pole Mission) — **Pittsburgh, PA**

Autonomous Systems Engineer (Profs. Red Whittaker & Heather Jones); NASA Ames & Astrobotic collaboration — *Jan 2025 – May 2025*

- Optimized the NASA cFS Data Storage Application for time-indexed telemetry (2MB time-queried packet transfers to Earth) and authored a ground-side Python CCSDS Space Packet Protocol parser for rover data decoding
- Diagnosed and patched a mission-blocking memory leak and SIGABRT segfault in the Data Storage Application, restoring the rover's ability to persist and downlink scientific telemetry

FIRST Robotics Competition Team #7539 Elev8 — **Mumbai, India**

Team Captain — *May 2021 – May 2024*

- Led 60-student team to win the **FRC Autonomous Award** by deploying cubic-spline/Bézier trajectory generation, EKF-based visual-inertial odometry, and tuned PID+feedforward controllers on real hardware, improving the team's autonomous score by 60%

Projects

LLM-Driven DAG-Based Multi-Robot Abstract Task Execution | *Python, AI2-THOR, Mistral*

- Designed a four-tier multi-agent motion planner (Prioritized Planning, CBS, PBS, BFS fallback with runtime collision guard) consuming LLM-generated DAGs over AI2-THOR scenes, with online task recovery for failed actions
- Achieved 1.50–2.34× makespan speedup and 100% lenient task completion on 34/36 trials across 9 prompt-scene pairs with up to 4 agents; Prioritized Planning alone resolved over 92% of plan calls excluding outliers

Flow-Matched Action Regularization for Quadruped Locomotion | *PyTorch, NVIDIA Isaac Sim, Unitree Go1*

- Trained a flow-matching velocity network to refine raw PPO locomotion actions toward physics-aware targets generated via short-horizon shooting
- Reduced fall rate by 46 pp (50% to 3.75%) and high-frequency action spectral energy by 19% on 50–500 N lateral push-recovery tests, adding only ∼6 ms inference overhead per 20 ms control step

Skills

Languages: Python, C, C++, Java, Bash
Robotics & ML Stack: ROS 2, NASA cFS, Gymnasium, PyTorch, TensorFlow, Keras, Eigen, Docker, Git/GitHub/GitLab, Linux
Simulation & CAD: SOFA, MuJoCo, NVIDIA Isaac, AI2-THOR, Blender API, SolidWorks, OnShape, Fusion 360
Technical Areas: MPPI & Model Predictive Control, Sampling-Based Optimization, Reinforcement Learning, Sim-to-Real Transfer, Motion Planning & MAPF, Vehicle & Soft-Robot Dynamics, State Estimation & Sensor Fusion, SLAM, Computer Vision, Kinematics